

HEALTHCARE DATA CLIENT

Power BI Requirements

Data Quality and Exception Reporting

Document Control	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Reporting Tool	Power BI
Batch Scale	Approximately 250,000 records per batch
Confirmed Results	2% exception rate; review turnaround reduced from 6 hours to 2.5 hours per file
Requirements status: Power BI reporting is confirmed. Exact historical dashboard page names, layouts, visual types, refresh schedule and DAX implementation are not retained. This document defines an evidence-safe portfolio reporting specification.	

1. Reporting Objective

Provide clear visibility into healthcare data-quality processing and exception outcomes so that the Healthcare Analyst can understand batch volume, identify records requiring review, investigate validation reasons and reconcile reporting to the automated validation output.

2. Reporting Scope

In Scope	Not Claimed
Processed batch volume	Exact historical dashboard layout
Record-level exception status	Exact historical visual types
2% observed Exception Rate	Per-rule historical counts not retained
Reason-code / validation-rule analysis where data supports it	Most common historical exception category
Validation categories	Healthcare-domain exception ranking
Review turnaround outcome	Unconfirmed refresh frequency or scheduling
Reconciliation to validation outputs	Unretained production architecture
Power BI reporting visibility	Clinical, compliance or patient-outcome reporting

3. Core Reporting Requirements

ID	Requirement	Trace
PBI-01	Display Records Processed for the governed batch population.	BR-01; KPI-01
PBI-02	Display distinct Exception Records requiring review.	BR-10, BR-11; KPI-02
PBI-03	Calculate Exception Rate using distinct Exception Records divided by Records Processed.	BR-14; KPI-03
PBI-04	Support analysis by validation rule or reason code when rule-level output is available.	BR-12; KPI-06
PBI-05	Represent the confirmed validation areas: duplicates, missing values, invalid dates, incorrect formats, invalid IDs, out-of-range values, inconsistent codes and cross-field mismatches.	BR-03 to BR-09
PBI-06	Allow record-level exception review using only source-supported review fields.	BR-11 to BR-13
PBI-07	Maintain traceability between dashboard results and governed validation outputs.	BR-17
PBI-08	Reconcile Power BI measures to processed, exception and validation-failure populations.	BR-18
PBI-09	Present confirmed review turnaround context: 6 hours before and 2.5 hours after per file.	Confirmed outcome
PBI-10	Avoid exposing confidential patient, provider or client information in public portfolio outputs.	Portfolio confidentiality control

4. KPI Requirements

Measure	Definition	Display Rule
Records Processed	Total records in the governed processed batch.	Show batch context; approximately 250,000 is confirmed scale, not a fixed target.
Exception Records	Distinct records failing one or more applicable rules.	Count each affected record once.
Exception Rate	Exception Records / Records Processed x 100.	Observed result is 2%; do not present 2% as a target or tolerance.
Validation Failures	Rule-level failures across records.	Keep separate from Exception Records because one record can fail multiple rules.
Exceptions by Reason	Exception records or failures grouped by governed reason information.	Do not fabricate historical category values.
Review Turnaround	Elapsed review time per file.	Confirmed comparison: 6 hours before, 2.5 hours after.
Turnaround Reduction	Baseline minus post-solution time.	3.5 hours per file, derived.
Turnaround Reduction %	Turnaround reduction / baseline x 100.	Approximately 58.3%, derived.

5. Proposed Portfolio Report Pages

Page-name boundary: The page names below organize the portfolio rebuild. They are not presented as the exact historical client dashboard page names.

Page	Purpose	Evidence-Safe Content
Data Quality Executive Overview	Summarize batch quality and review outcome.	Records Processed, Exception Records where source output is available, 2% Exception Rate, validation scope, review turnaround context.
Exception Analysis	Support investigation of records requiring review.	Exception status, reason code, validation rule, multi-failure logic and record-level review table where source-supported.
Validation Performance	Explain validation coverage and reconciliation.	Eight confirmed validation categories, processed population, validation-failure logic, exception reconciliation and turnaround outcome.

6. Executive Overview Requirements

Component	Requirement
Batch Volume	Show governed Records Processed.
Exception Rate	Show record-level Exception Rate using the governed formula.
Exception Context	Show Exception Records only when the underlying governed population is available.
Validation Coverage	Show the eight confirmed validation categories without inventing failure counts.
Turnaround	Show 6 hours before and 2.5 hours after per file as confirmed outcome context.
Outcome Interpretation	Describe the change as review turnaround reduction, not productivity, labor-cost or clinical improvement.

7. Exception Analysis Requirements

Component	Requirement
Exception Table	Provide review-level visibility using controlled, source-supported fields.
Reason Analysis	Group by governed reason code or validation rule where data exists.
Multi-Failure Handling	Allow one record to retain multiple rule failures without double-counting Exception Records.
Filtering	Permit filtering by available batch, validation rule, reason or source-supported domain where those fields exist.
No Unsupported Ranking	Do not claim a top exception category, highest-risk domain or severity ranking without evidence.
Traceability	Maintain a source record reference or equivalent governed linkage in the controlled implementation.

8. Validation Performance Requirements

Component	Requirement
Rule Coverage	Represent all implemented validation rule families.
Record vs Failure Grain	Keep record-level exception measures separate from rule-level failure measures.
Reconciliation	Support validation that every Exception Record has at least one failure.
Reason Consistency	Use governed rule/reason classifications consistently across reporting.
Batch Context	Retain the processing context needed to reconcile results.
Historical Boundary	Do not create unsupported trends or per-rule performance values.

9. Data Model Requirements

Requirement	Design Control
Record grain	Maintain a governed processed-record population for Records Processed and Exception Records.
Failure grain	Maintain rule-level detail where multiple failures per record need analysis.
Relationship	Link rule-level failures to the affected processed record without multiplying record-level KPIs.
Reason dimension	Use governed validation rule or reason attributes when available.
Batch context	Retain a batch or source reference where available for reconciliation.
Healthcare domain	Use domain classification only when explicitly supported by the source data.
Confidentiality	Do not publish real PHI or confidential client values in portfolio datasets or screenshots.

10. Interaction Requirements

Interaction	Requirement
Filter context	Measures must respond consistently to supported report filters.
Drilldown	Reason or rule drilldown may be used only when underlying rule-level detail exists.
Record review	Detailed views should expose only fields required for controlled review.
Cross-filtering	Interactions must preserve the distinction between record-level and rule-level grain.
Reset / default view	Portfolio rebuild may provide a clean default state without claiming a historical client configuration.

11. Data & Measure Validation

Test	Expected Result
Records Processed reconciliation	Power BI total matches the governed processed population.
Exception Records reconciliation	Dashboard distinct exception count matches the governed exception population.
Exception Rate calculation	Exception Records / Records Processed x 100.
Multi-failure test	A record with multiple failures counts once in Exception Records.
Reason analysis	Rule-level detail reconciles to the validation-failure output.
Turnaround calculation	6 hours to 2.5 hours equals 3.5 hours reduction and approximately 58.3% reduction.
Filter test	Record-level measures remain valid under supported filter context.

12. Reporting Acceptance Criteria

ID	Acceptance Condition
AC-PBI-01	Core record-level measures reconcile to governed validation outputs.
AC-PBI-02	Exception Rate uses distinct exception records and the governed denominator.
AC-PBI-03	Multiple validation failures do not inflate record-level exception metrics.
AC-PBI-04	Reason-code analysis uses available governed failure detail.
AC-PBI-05	The 2% result is labeled as observed, not as a target.
AC-PBI-06	The 6-hour to 2.5-hour comparison is presented as review turnaround per file.
AC-PBI-07	No unsupported healthcare findings, thresholds or clinical conclusions are displayed.
AC-PBI-08	Public portfolio reporting contains no PHI or confidential client values.

13. Requirements Traceability

Reporting Area	Upstream Artifact	Downstream Artifact
KPIs	KPI Catalogue	Data Model & DAX Measures
Validation categories	Data Quality Rule Catalogue	Dashboard; Testing
Exception analysis	Exception Analysis	Dashboard; Findings
Workflow context	Automated Validation Workflow	Testing; Final Report
Record and failure grain	Data Dictionary	Data Model & DAX Measures
Confirmed outcomes	KPI Catalogue	Findings; Final Report